Machine Discovery of Mathematical Conjectures and Structures

DSA5210 Session 09 —AI for Mathematics

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Relations among numbers

PSLQ: finding a relation among numbers

PSLQ detects **integer relations**. Given reals $x_1, \ldots, x_n$ to high precision, it returns integers $a_1, \ldots, a_n$ (not all zero) with

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = 0,$$

or certifies that none exists below a height bound.

- Ferguson–Forcade (1979); the stable PSLQ is Ferguson–Bailey–Arno (1999); related to LLL (Lenstra–Lenstra–Lovász) lattice reduction.
- Feed it a constant's value plus candidate terms (π, logs, ζ values, powers); it returns an exact-looking identity.
- Named one of the "top ten algorithms of the century" (2000).

The identity is a candidate: true to the computed precision, still to be proved.

An integer relation is a short vector in a lattice

Embed the numbers: take $\mathbf{b}_i = (\mathbf{e}_i,\ Nx_i)$, where $\mathbf{e}_i = (0, \ldots, 1, \ldots, 0)$ is the i-th unit vector and N is large. Each $\mathbf{b}_i$ is an $(n{+}1)$-vector, so

$$\sum_i a_i \mathbf{b}_i = \big(\underbrace{a_1, \ldots, a_n}_{\text{from the } \mathbf{e}_i},\ \underbrace{N \textstyle\sum_i a_i x_i}_{\text{last coordinate}} \big).$$

If $\sum_i a_i x_i \approx 0$ with small a_i, this lattice vector is **short**.

So **an integer relation** $=$ **a short vector in this lattice**, and the goal is to find the shortest one. That is what lattice reduction (LLL) does.

LLL: reduce the basis

Input a lattice basis $\mathbf{b}_1, \ldots, \mathbf{b}_n$.

Output a basis of the same lattice with a short first vector, $\|\mathbf{b}_1\| \le 2^{(n-1)/2}\lambda_1$ (λ_1 = true shortest length); $\mathbf{b}_1$ is the candidate relation.

Minimizes the potential $D = \prod_k \prod_{i \le k} \|\mathbf{b}_i^*\|^2$ (product of the partial Gram determinants).

($\mathbf{b}_i^*$: Gram–Schmidt vectors; $\mu_{ij} = \langle \mathbf{b}_i, \mathbf{b}_j^* \rangle / \langle \mathbf{b}_j^*, \mathbf{b}_j^* \rangle$.)

A descent on D, alternating two moves until neither applies:

- **Size-reduce**: $\mathbf{b}_i \leftarrow \mathbf{b}_i - \lfloor \mu_{ij} \rceil \mathbf{b}_j$ until $|\mu_{ij}| \le \frac{1}{2}$ — near-orthogonal to earlier vectors; D fixed.
- **Lovász swap**: if $\|\mathbf{b}_i^*\|^2 < (\delta - \mu_{i,i-1}^2)\|\mathbf{b}_{i-1}^*\|^2$ ($\delta = \frac{3}{4}$), swap $\mathbf{b}_{i-1} \leftrightarrow \mathbf{b}_i$ — the shorter vector moves up; each swap shrinks D by a factor $< \delta$, so the loop stops.

PSLQ: input, output, objective

Input reals $x_1, \ldots, x_n$ to high precision.

Output a nonzero integer vector $\mathbf{a}$ with $\sum_i a_i x_i = 0$; or a bound M with every relation $\|\mathbf{a}\| \geq M$ (none below height M).

Objective the smallest integer relation, $\min \|\mathbf{a}\|$ over $0 \neq \mathbf{a} \in \mathbb{Z}^n$ with $\sum_i a_i x_i = 0$ — the same shortest vector LLL seeks.

The difference from LLL is in *how*: PSLQ reduces the real numbers x_i directly, kept to high precision, instead of building LLL's integer lattice $\mathbf{b}_i = (\mathbf{e}_i, N x_i)$. So there is no large multiplier N to pick or inflate, and the arithmetic stays numerically stable.

PSLQ: the matrix $H_{\mathbf{x}}$

Every integer relation lies in the **relation space** $\mathbf{x}^{\perp} = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v} \cdot \mathbf{x} = 0\}$. Normalize $|\mathbf{x}| = 1$ and set $s_k = \sqrt{x_k^2 + \cdots + x_n^2}$ (so $s_1 = 1$). These define the $n \times (n-1)$ matrix

$$(H_{\mathbf{x}})_{jj} = \frac{s_{j+1}}{s_j}, \quad (H_{\mathbf{x}})_{ij} = -\frac{x_i x_j}{s_j s_{j+1}} \ (i > j), \quad (H_{\mathbf{x}})_{ij} = 0 \ (i < j),$$

whose $n-1$ columns are orthonormal and orthogonal to $\mathbf{x}$ — a basis of $\mathbf{x}^{\perp}$.

Certificate (Ferguson–Bailey–Arno): for any invertible integer matrix A and orthogonal Q with $AH_{\mathbf{x}}Q$ lower-trapezoidal (zero above the diagonal), every relation $\mathbf{m}$ satisfies $|\mathbf{m}| \geq 1/\max_j |(AH_{\mathbf{x}}Q)_{jj}|$.

PSLQ: the loop

Keep an integer matrix A (with its inverse A^{-1}) and an orthogonal Q; write $H = AH_{\mathbf{x}}Q$. Fix a parameter $\gamma > 2/\sqrt{3}$, then repeat:

Reduce subtract rounded integer multiples of earlier rows ($A \leftarrow DA$, D integer) until $|H_{ij}| \le \frac{1}{2}|H_{jj}|$ for $i > j$.

Exchange pick r maximizing $\gamma^r |H_{rr}|$; swap rows $r, r{+}1$ of A.

Corner the swap breaks the lower-trapezoidal form; a plane (Givens) rotation $Q \leftarrow QP$ on columns $r, r{+}1$ restores it (the "LQ").

Stop when $\mathbf{x}A^{-1}$ has a zero entry: that column of A^{-1} is the relation $\mathbf{m}$. Until then $1/\max_j |H_{jj}|$ lower-bounds every relation — the "no relation below height M" certificate.

A worked discovery: the BBP formula for π

The Bailey–Borwein–Plouffe (BBP) formula:

$$\pi = \sum_{k=0}^{\infty} \frac{1}{16^k} \left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6} \right)$$

- Found numerically in 1995 (Plouffe) by an integer-relation search; **proved** after the discovery, and published with the proof in 1997.
- Gives the n-th hexadecimal digit of π — and the $4n$-th binary digit — without the earlier digits, in little memory.
- Base 16, not base 10: cheap decimal-digit extraction is still open.

Discovered from the numbers, proved afterward.

PSLQ: rediscovering the BBP formula

Feed PSLQ the eight building blocks $S_j = \sum_{k \geq 0} 1/(16^k(8k+j))$, with no prior knowledge of the coefficients:

```
import mpmath as mp;  mp.mp.dps = 100
S = [mp.nsum(lambda k, j=j: 1/(mp.mpf(16)**k*(8*k+j)), [0, mp.inf])
     for j in range(1, 9)]              # S1 .. S8
rel = mp.pslq([mp.pi] + S, maxcoeff=10**6)
```

Output:

```
rel = [1, -4, 0, 0, 2, 1, 1, 0, 0]   ==  published BBP
  ->  pi = 4*S1 - 2*S4 - S5 - S6        (rel. error vs pi: 9e-102)
spigot: pi hex digits at offset 999 = 349F1C   (matches mpmath)
PSLQ finds nothing below 16 digits, then locks on (height 4)
```

The exact formula comes back from the numbers alone; the spigot step then extracts the n-th hexadecimal digit of π directly.

From computed numbers to a closed form: OEIS, LIReC

- **OEIS**, the On-Line Encyclopedia of Integer Sequences (oeis.org; Sloane; over $390{,}000$ sequences) — type the first terms of a computed sequence; it returns matches with conjectured formulas, generating functions, recurrences, and references.
- **LIReC**, the Library of Integer RElations and Constants (github.com/RamanujanMachine/LIReC) — a database of constants and a hypergraph of PSLQ-found relations; a 2024 sweep reported 75 previously unknown connections, including new π–e relations.

LIReC: recovering an integer relation

Three constants to 190 digits — $\zeta(3)$, $\zeta(5)$, and $x = 2 + 3\zeta(3) - 7\zeta(5)$ built from them — handed to LIReC's PSLQ core:

```
from LIReC.lib.pslq_utils import PreciseConstant as C, check_consts
import mpmath as mp;  mp.mp.dps = 200
z3 = C(mp.zeta(3), 190, 'zeta3')
z5 = C(mp.zeta(5), 190, 'zeta5')
x  = C(2 + 3*mp.zeta(3) - 7*mp.zeta(5), 190, 'x')
check_consts([z3, z5, x], degree=1)
```

Output (run on the cluster, mu012):

```
[ x - 3*zeta3 + 7*zeta5 - 2 = 0   (188) ]
```

The integer relation comes back exactly, valid to 188 digits. Put an unknown constant in place of x and the same call is a discovery.

Formulas from data

Symbolic regression: a formula from data

Input numeric pairs (x_i, y_i) and an operator set $\{+, -, \times, \div, \exp, \log, \sqrt{}\}$.

Output a *Pareto front* of closed-form formulas f — for each complexity, the best-fitting one.

Minimizes two quantities jointly — the data misfit $\sum_i \big(f(x_i) - y_i\big)^2$ and the expression complexity (number of operators/nodes in the formula). They trade off, so the answer is a front, not one formula.

Algorithm: genetic programming over expression trees — keep a population of candidate formulas, repeatedly mutate and recombine them, and select by a fitness that rewards low error and small size.

- **PySR** (Cranmer, 2023) is this implementation; AI Feynman and SINDy (sparse identification of nonlinear dynamics) are relatives.
- It returns the functional form — no proof, no error bound; a fitted constant is pinned afterward by an integer-relation step.

Partition asymptotics

Let $p(n)$ count the partitions of n — the ways to write n as a sum of positive integers, order ignored. Its generating function is

$$\sum_{n} p(n) x^n = \prod_{k \geq 1} (1 - x^k)^{-1},$$

which lets $p(n)$ be computed exactly in integer arithmetic, with no rounding, up to $n = 20{,}000$.

The Hardy–Ramanujan formula (1918) is

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi \sqrt{\tfrac{2n}{3}}\right).$$

Discovery task: compute $p(n)$, take logarithms, run symbolic regression on $(n, \log p(n))$.

Euler's pentagonal number theorem

The product $\prod_{k\geq 1}(1-x^k)$ — the reciprocal of the partition generating function — expands to an almost-empty signed sum:

$$\prod_{k\geq 1}(1-x^k) = \sum_{j=-\infty}^{\infty} (-1)^j\, x^{j(3j-1)/2} = 1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + \cdots$$

Non-zero terms sit only at the generalized pentagonal numbers $\frac{k(3k\pm 1)}{2} = 1, 2, 5, 7, 12, 15, \ldots$, with coefficient $(-1)^k$ on the k-th pair.

The coefficient of x^n counts the partitions of n into distinct parts with sign $(-1)^{\#\text{parts}}$; even and odd cancel in pairs (Franklin's bijection, 1881), leaving ± 1 only at a pentagonal n.

Matching x^n coefficients in $\left(\sum_n p(n)x^n\right)\prod_k(1-x^k) = 1$ gives the exact recurrence

$$p(n) = p(n{-}1) + p(n{-}2) - p(n{-}5) - p(n{-}7) + p(n{-}12) + \cdots,$$

an integer sum over the $O(\sqrt{n})$ pentagonal numbers below n.

Symbolic regression recovers the Hardy–Ramanujan form

Run PySR on $(n, \log p(n))$ with no form supplied — four random seeds in parallel on the CPU cluster, minutes each. Three of the four independently recover (complexity 11, loss $\sim 10^{-10}$):

$$\log p(n) \approx 2.565099\,\sqrt{n - 0.349} - \log n - 1.9355.$$

- $\sqrt{n}$ coefficient 2.565099 vs $\pi\sqrt{2/3} = 2.5650997$; constant -1.9355 vs $-\log(4\sqrt{3}) = -1.93560$ — 5–7 significant figures.
- The -0.349 inside the root is a sub-leading correction, beyond the crude 1918 leading term.
- The proof of the formula is separate from the fit: the circle method, the modular transformation of the generating function, Rademacher's exact series (1937).

The improved asymptotic: Rademacher's leading term

The 1918 formula is only the crude leading term. The first term of Rademacher's exact series (1937) is

$$p(n) \sim \frac{\sqrt{12}}{24n-1}\left(1-\frac{1}{\mu}\right)e^{\mu}, \qquad \mu=\frac{\pi}{6}\sqrt{24n-1},$$

with relative error $\sim 10^{-14}$ by $n=2000$, where the crude term is still 1% off.

The fitted -0.349 shift is built from two refinements inside this term:

- $24n-1=24\big(n-\frac{1}{24}\big)$, so $\mu=\pi\sqrt{\frac{2}{3}}\sqrt{n-\frac{1}{24}}$: the root is shifted by $\frac{1}{24}$.
- the factor $1-\frac{1}{\mu}$ contributes a further $O(1/\sqrt{n})$ term.

Folded into a single $\sqrt{}$-shift, the two give $\frac{1}{24}+\frac{3}{\pi^2}=0.346$, against the fitted 0.349.

The Pareto front for $\log p(n)$

Symbolic regression returns the best formula at each complexity, not a single answer. For this run:

complexity	best formula	loss
4	$2.460\,\sqrt{n}$	7.5
9	$2.5652\,\sqrt{n} - \log n - 1.952$	1.0×10^{-5}
11	$2.565099\,\sqrt{n - 0.349} - \log n - 1.9355$	2.7×10^{-10}

Complexity 11 is the knee: past it, extra terms cut the loss by only $\sim 10^{-10}$. The complexity-4 row, $2.460\,\sqrt{n}$, is the leading term by itself — the search reaches the full form only after it adds the $-\log n$ term.

Recovering the constant by least squares

With the form assumed, the coefficients follow from a linear least-squares fit of $a\sqrt{n} + b\log n + c$ to $(n, \log p(n))$, $50 \le n \le 20{,}000$:

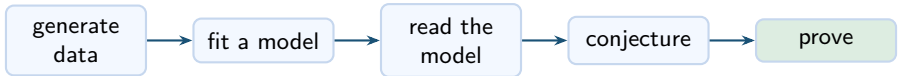
fit	a	ground truth	error
$a\sqrt{n} + b\log n + c$	2.56491	$\pi\sqrt{2/3} = 2.56510$	rel. 7×10^{-5}
$a\sqrt{n}$ only	2.460	same	4%

The full form pins $\pi\sqrt{2/3}$ to five significant figures; the leading term alone is 4% off — the same 2.460 the symbolic-regression front shows at complexity 4. The $-\log n$ term, from the $1/n$ prefactor in the Hardy–Ramanujan formula, is what fixes the constant.

Rules from labelled data

The pipeline: fit a model, then read it

Supervised learning can surface a hidden rule. The pipeline is the one Davies and collaborators used for knot theory and Kazhdan–Lusztig polynomials (Nature 2021).



- Exact labels from a computer algebra system — no measurement noise.
- Read the model through feature importance or saliency — which inputs move the output.
- The machine stops at the conjecture; a human proves it.

Decision tree: a rule from labelled data

Input	labelled triples — each (a, b, c) tagged appears (1) or absent (0), presented as raw spins or as difference features.
Output	a tree of one-variable threshold tests ("is $a + b - c < 0$?"), a label at each leaf; each root-to-leaf path is an explicit rule.
Minimizes	at each node, the weighted Gini impurity of the two children — $\text{Gini} = 1 - p^2 - (1-p)^2 = 2p(1-p)$ for a node with appears-fraction p, and 0 at a pure leaf. Equivalently, it maximizes the impurity drop.

Algorithm: greedily pick the single feature and threshold that minimizes that weighted child impurity; recurse on each branch until the leaves are pure (or a depth cap). The model is white-box — the rule reads off the tree.

Two other readouts of a fitted model: inductive logic programming returns a logical clause; a black-box classifier is read by saliency, the gradient of the output with respect to each input.

Clebsch–Gordan selection rules — the toy

For $SU(2)$, the tensor product of spin-a and spin-b decomposes as

$$V_a \otimes V_b = \bigoplus_{c=|a-b|}^{a+b} V_c,$$

in integer steps; each multiplicity is 0 or 1. It is 1 exactly when

$$|a-b| \le c \le a+b \quad \text{and} \quad a+b+c \in \mathbb{Z}$$

— a triangle inequality plus an integrality condition.

Rediscover it: label all triples (a, b, c) with $a, b \le 10$ from an independent computation of the multiplicity (about $11{,}000$ triples, 30% positive, so "always absent" already scores 0.70), and train a decision tree.

Raw spins vs differences

The outcome turns entirely on how each triple is presented to the tree.

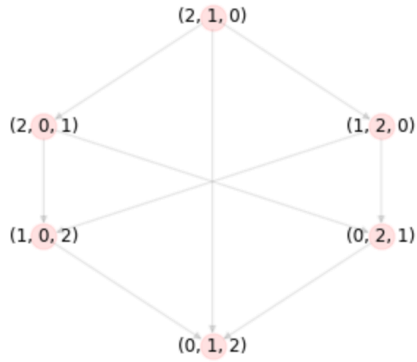
- **Raw** (a, b, c)**:** a depth-12 tree scores 0.58, below the 0.70 baseline — the integrality condition is not a single-coordinate threshold.
- **Differences** $(\, a{+}b{-}c,\ c{-}|a{-}b|,\ [\, a{+}b{+}c \in \mathbb{Z}\,]\,)$: each part of the rule is now one threshold; a depth-4 tree is exact, and trained on $a, b \le 6$ it still scores 1.00 on larger spins.

The depth-4 tree reads back as the rule:

$$\text{appears} \iff a + b + c \in \mathbb{Z} \ \wedge\ c \ge |a - b| \ \wedge\ c \le a + b.$$

The symmetric group and the Bruhat graph

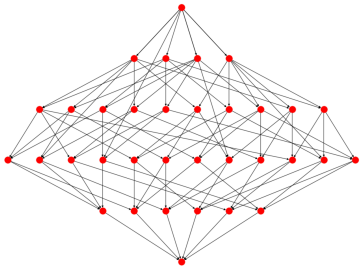
- The symmetric group S_N is all permutations of $\{0, 1, \ldots, N-1\}$, written in string notation: 2031 is the permutation $0 \mapsto 2,\ 1 \mapsto 0,\ 2 \mapsto 3,\ 3 \mapsto 1$.
- Bruhat graph: one node per permutation; an edge joins two permutations that differ by a transposition — a swap of two of the values.
- The length $\ell(x) = \#\{\, i < j : x(i) > x(j) \,\}$ counts inversions. It gives each node a height and orients every edge downward, from longer to shorter.



The ordered Bruhat graph of S_3: top is the reverse permutation 210, bottom is the identity 012.

Bruhat order and the interval $[x, y]$

- Bruhat order: $x \leq y$ when there is a downward path from y to x in the Bruhat graph. The identity is the least element, the reverse permutation w_0 the greatest.
- The interval $[x, y]$ is the set of permutations z with $x \leq z \leq y$, together with the edges among them — a directed graph in its own right.
- Erasing the vertex names leaves the *unlabelled* interval: the shape of that graph alone.



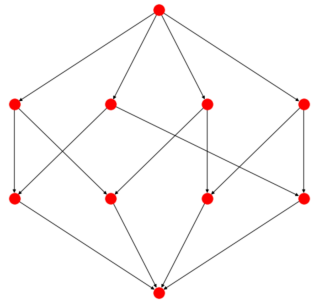
An interval $[03214,\ 34201]$ inside S_5.

Kazhdan–Lusztig polynomials

- Kazhdan and Lusztig (1979) attached to each pair $x \leq y$ a polynomial $P_{x,y}(q)$ with integer coefficients.
- The definition is inductive on length — computing $P_{x,y}$ may call on $P_{u,v}$ for many pairs below. It is awkward by hand, but a computer generates billions.
- Two facts already tie the polynomial to the graph alone:
  - $P_{x,y} \neq 0$ exactly when $x \leq y$;
  - $P_{x,y} = 1$ exactly when the unoriented interval $[x, y]$ is regular (every vertex has the same degree).
- Why they matter: $P_{x,y}$ records the local intersection cohomology of Schubert varieties (Kazhdan–Lusztig 1980) and composition multiplicities of Verma modules in representation theory.
- Small values: $1, \quad 1+q, \quad 1+q^2, \quad 1+3q+q^2$.

The combinatorial invariance conjecture

- Conjecture (Lusztig and Dyer, independently, 1980s): $P_{x,y}$ depends only on the isomorphism type of the unlabelled interval $[x, y]$ — not on which permutations x and y are.
- Example in S_4: the intervals $[0213, 2301]$ and $[1032, 3120]$ are different, yet both are the same directed graph, the "4-crown", and both have $P_{x,y} = 1 + q$.
- Known when x is the identity; open in general.



The "4-crown". It occurs as two different intervals in S_4, both with KL polynomial $1 + q$.

Graph neural networks

- A graph neural network computes on a graph directly. Each node holds a vector of numbers, started from its features — here its in-degree and out-degree.
- **Message passing:** in each round every node replaces its vector by combining its own with an aggregate (sum or mean) of its neighbours' vectors, through a small learned function. Four rounds were used, so each node ends up summarising its four-hop neighbourhood.
- **Readout:** the node vectors are pooled into one graph-level output — here the predicted coefficients of $P_{x,y}$.
- One set of weights is shared across all nodes and all graphs, so a single network handles intervals of any size and depends only on the connectivity, not on the node names.

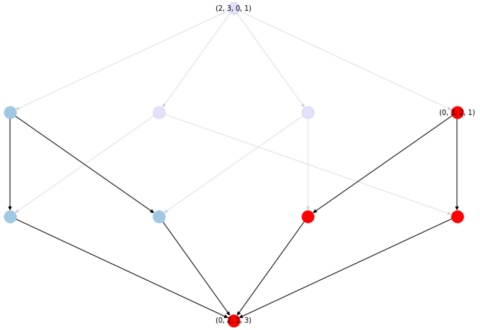
Learning the polynomial from the graph

- **Data:** every interval in the symmetric groups up to S_9 — about 2.9 million in S_9, reduced to $24{,}322$ distinct interval graphs; split $80/20$ into train and test.
- **Input:** the unlabelled interval graph, each node carrying its in-degree and out-degree; edge labels (which transposition) are withheld.
- **Model:** a message-passing graph neural network, four rounds, predicting each coefficient of q, q^2, q^3, q^4 as a separate classification.
- **Result:** the coefficients were predicted with about 98% accuracy (Williamson, 2023) — evidence that $P_{x,y}$ is a function of the graph alone.

Geordie Williamson — representation theorist, University of Sydney; found the hypercube decomposition.

Saliency and the hypercube decomposition

- Gradient saliency ranks each edge by its effect on the prediction. The extremal reflections — those moving the first or last position — stand out far beyond chance.
- Williamson turned that into a decomposition of every interval, along its extremal reflections, into
  - a hypercube, from one set of extremal edges;
  - a subgraph isomorphic to an interval in S_{N-1}.
- From it $P_{x,y}$ is computed directly from the graph — checked on all intervals up to S_9 (over a million); conjectured in general, which would imply combinatorial invariance for S_N (Blundell et al., 2022).



Interval $[(0,2,1,3),(2,3,0,1)]$ in S_4: hypercube piece (blue), interval-in-S_{N-1} piece (red).

Littlewood–Richardson and Kronecker coefficients

Two more multiplicities, both central in algebraic combinatorics:

- Littlewood–Richardson $c^{\lambda}_{\mu\nu}$ — V_λ inside $V_\mu \otimes V_\nu$ for GL_n; the Schur structure constants.
- Kronecker $g(\lambda, \mu, \nu)$ — S^λ inside $S^\mu \otimes S^\nu$ for the symmetric group; no combinatorial rule is known (Murnaghan's problem, open since 1938).

The value is hard to compute: #P-complete for Littlewood–Richardson, #P-hard for Kronecker — #P is the class of counting problems ("how many solutions", e.g. counting the satisfying assignments of a Boolean formula), at least as hard as NP — and for Kronecker even deciding which coefficients are nonzero is NP-hard. But the *support* — which are zero — is a classification problem, and a computer algebra system supplies exact labels in any quantity. Classifiers predict it well and sharpen as n grows (the ML4AlgComb — machine learning for algebraic combinatorics — datasets, Kvinge et al. 2025).

The same pipeline in the Langlands program

Lanard and Mínguez (2024) used this pipeline to find an algorithm for Aubert–Zelevinsky duality — an involution on the irreducible representations of a p-adic group — on SO_{2n+1} and Sp_{2n}, extending the Mœglin–Waldspurger algorithm, the known combinatorial rule for that duality on GL_n.

- Predicting the whole answer failed, so they predicted one piece of the combinatorial data at a time — the smallest of the "segments" indexing the representation — and read the formula off that.
- They calibrated the saliency on GL_n, where the algorithm is known, before trusting it on the classical groups, where it was not.

Conjecture generators

Graffiti: conjecturing inequalities between invariants

Graffiti (Fajtlowicz, mid-1980s) proposes inequalities among graph invariants.

- Keep a database of graphs; compute ~ 60 invariants (independence number, domination number, average distance, matching number, eigenvalues).
- Propose an inequality whenever it holds for every graph in the database.
- The Dalmatian heuristic keeps a new conjecture only if it is sharper on some graph than all kept so far.

Circulated as "Written on the Wall"; about eighty papers followed, by Chung, Erdős, Lovász, Seymour and others.

The Ramanujan Machine: continued fractions for constants

Raayoni et al. (*Nature* 590, 2021) generate conjectured *polynomial continued fractions* (PCFs) for fundamental constants — π, e, Catalan's constant G, and zeta values $\zeta(2) = \pi^2/6$, $\zeta(3)$:

$$x = a_0 + \cfrac{b_1}{a_1 + \cfrac{b_2}{a_2 + \cfrac{b_3}{a_3 + \ddots}}}, \qquad a_n, b_n \in \mathbb{Z}[n]$$

— the partial numerators b_n and denominators a_n are polynomials in n with integer coefficients.

- **Input:** a target constant, and bounded families of integer polynomials a_n, b_n.
- **Output:** an identity "constant = PCF", matched numerically — a conjecture, with no proof attached.

The Ramanujan Machine: how it searches

Both methods match numerical values only — no proof, no assumed structure.

MITM-RF (Meet-in-the-Middle):

- Input: the target constant; a library of candidate PCFs.
- Evaluate both sides to low precision; hash one side, look up the other.
- Re-check every hit at high precision to discard false matches.

Descent&Repel (gradient descent): minimise the gap between a PCF's value and the target over the integer coefficients; a "repel" step pushes away from solutions already found, to surface new ones.

Matches are tested to up to 2000 digits: a chance match in a 10^9 search at 50-digit accuracy has probability below 10^{-40}.

The Ramanujan Machine: $\zeta(3)$ and Catalan's constant

Extending the search produced previously-unknown continued fractions for $\zeta(3)$ (Apéry's constant) and Catalan's constant $G = 0.91596\ldots$:

$$\frac{12}{7\zeta(3)} = 2 - \cfrac{16 \cdot 1^6}{36 - \cfrac{16 \cdot 2^6}{160 - \cfrac{16 \cdot 3^6}{434 - \ddots}}}$$

$$\frac{2}{2G - 1} = 3 - \cfrac{6 \cdot 1^3}{13 - \cfrac{8 \cdot 2^3}{29 - \cfrac{10 \cdot 3^3}{51 - \ddots}}}$$

Both were matched from the numerical value alone, with no assumed structure, and were unproven when published; the Catalan-constant conjectures are still open.

The Ramanujan Machine: example formulas and status

Two formulas the machine found (both later proved):

$$\frac{e}{e-2} = 4 - \cfrac{1}{5 - \cfrac{2}{6 - \cfrac{3}{7 - \ddots}}}, \qquad \frac{4}{3\pi - 8} = 3 - \cfrac{1 \cdot 1}{6 - \cfrac{2 \cdot 3}{9 - \cfrac{3 \cdot 5}{12 - \ddots}}}.$$

- The program finds the identity; it cannot prove it.
- Yamamoto (2024) proved 38 of the conjectures: each PCF reduces to a second-order linear difference equation, solved via a differential equation or Petkovšek's algorithm; 31 were generalised to wider families.
- Many others, including Catalan's-constant conjectures, remain open.

Murmurations of elliptic curves

Elliptic curves over $\mathbb{Q}$

- An **elliptic curve** over the rationals is the solution set of

$$E:\ y^2 = x^3 + ax + b, \qquad a, b \in \mathbb{Q},$$

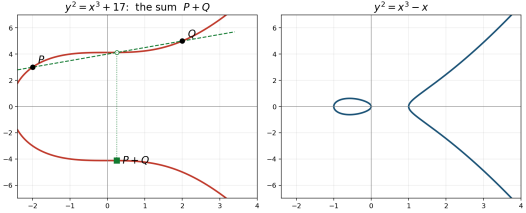
  together with one point O "at infinity".
- **Smoothness:** require the discriminant $\Delta = -16(4a^3 + 27b^2) \neq 0$, so the curve has no cusp or self-crossing.
- $E(\mathbb{Q})$ is the set of rational points (x, y) on E, plus O. The basic questions: how many are there, and how are they related?
- Over $\mathbb{R}$ the curve is one branch, or one branch plus a closed oval — $y^2 = x^3 - x$ has both.

The group law: chord and tangent

$E(\mathbb{Q})$ is an abelian group. A line meets the cubic in three points (with multiplicity); the operation $P + Q$ is built from that:

- **Sum.** The line through P, Q meets E in a third point R; reflect R across the x-axis to get $P + Q$.
- **Double.** For $2P$, use the *tangent* at P in place of the chord, then reflect.
- **Identity.** O, the point at infinity where all vertical lines meet, is neutral: $P + O = P$.
- **Inverse.** $-P$ is the reflection of P; the line through them is vertical, so its third point is O and $P + (-P) = O$.

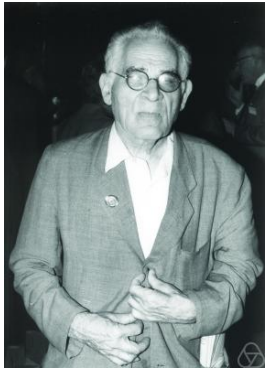
Commutative and associative; $nP = P + \cdots + P$ builds infinitely many points from a few.



Left: the sum $P + Q$ on $y^2 = x^3 + 17$ via the chord and the reflection. Right: the two-component locus $y^2 = x^3 - x$.

Mordell's theorem and the rank

- **Mordell (1922):** $E(\mathbb{Q})$ is a finitely generated abelian group,
$$E(\mathbb{Q}) \;\cong\; \mathbb{Z}^r \;\oplus\; E(\mathbb{Q})_{\mathrm{tors}},$$
with $E(\mathbb{Q})_{\mathrm{tors}}$ finite.
- The **rank** $r \geq 0$ is the number of independent infinite-order points that generate, by the group law, all but finitely many rational points.
- Rank 0: finitely many rational points; rank ≥ 1: infinitely many.
- Computing the rank exactly is hard — no algorithm is proven to terminate for every curve.
- But it can be *predicted* from the a_p with high accuracy — statistics, not proof.



Louis Mordell (1888–1972): proved $E(\mathbb{Q})$ finitely generated, 1922. Sadleirian Chair, Cambridge.

From $E(\mathbb{Q})$ to $E(\mathbb{F}_p)$: why count mod p

- **The rank is hard to reach directly.** It lives over $\mathbb{Q}$, the rational points can be infinite, and no algorithm is proven to compute it for every curve.
- **Reduce mod p.** For a good prime $p \nmid \Delta$, reduce the coefficients a, b mod p: the result is an elliptic curve over the finite field $\mathbb{F}_p = \{0, \ldots, p-1\}$. Now x, y each take only p values, so the points, plus O, form a *finite* group $E(\mathbb{F}_p)$ —a directly computable count.
- **The link.** Reducing coordinates is a group homomorphism $E(\mathbb{Q}) \to E(\mathbb{F}_p)$ (good p): each rational point maps to a mod-p point. So $E(\mathbb{Q})$ is reflected in the finite groups, and the rank shows up as a *bias* in the counts N_p —more rational points, more points mod p on average. The Birch–Swinnerton-Dyer conjecture makes this link exact.

a_p and the L-function

- For each good prime, the point count gives

$$N_p = \#E(\mathbb{F}_p), \qquad a_p = p + 1 - N_p,$$

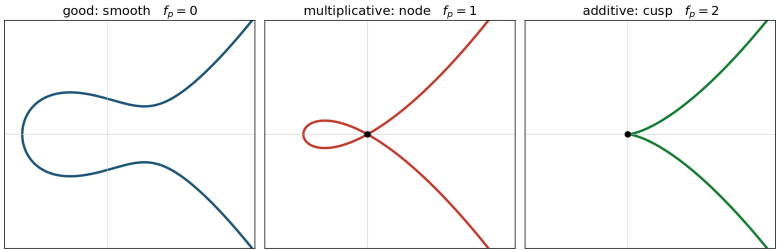
  the **Frobenius trace** — how far N_p strays from the expected $p+1$. Hasse: $|a_p| \le 2\sqrt{p}$.
- The (a_p) assemble into the L-**function**

$$L(E,s) = \prod_{p \text{ good}} \left(1 - a_p p^{-s} + p^{1-2s}\right)^{-1} = \sum_{n \ge 1} a_n n^{-s}, \quad \mathrm{Re}(s) > \tfrac{3}{2}.$$

- By the modularity theorem (Wiles; Taylor–Wiles; Breuil–Conrad–Diamond–Taylor) $L(E,s)$ extends to all of $\mathbb{C}$. Its center $s = 1$ is where the arithmetic concentrates.

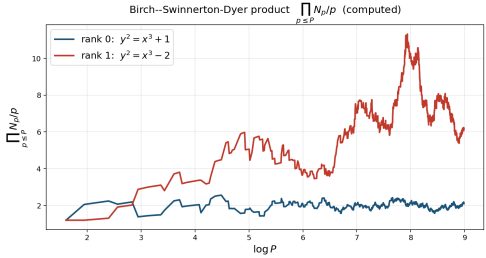
Reduction types and the conductor

- At a bad prime $p \mid \Delta$ the reduced curve is singular: a **node** (two tangent lines) is *multiplicative* reduction, a **cusp** (one) is *additive*; good reduction is smooth.
- The **conductor** is $N(E) = \prod_p p^{f_p}$ with $f_p = 0$ (good), 1 (mult.), 2 (additive, $p \geq 5$).
- At $p = 2, 3$ a wild term can raise it ($f_2 \leq 8$, $f_3 \leq 5$); in general $f_p = v_p(\Delta_{\min}) + 1 - m_p$ ($m_p = \#$components of the special fibre; Ogg, via Tate's algorithm).
- N square-free $\iff$ E semistable. The conductor orders curves by size and sets the murmuration bands.



The Birch–Swinnerton-Dyer conjecture, found by computer

- Birch and Swinnerton-Dyer, on the **EDSAC-2** at Cambridge (1958–62), computed N_p and studied $\displaystyle\prod_{p\le P}\frac{N_p}{p}$, finding it grows like $C(\log P)^{\,r}$ — higher rank, more points mod p.
- **Conjecture.** $\operatorname{rank} E(\mathbb{Q}) = \operatorname{ord}_{s=1} L(E,s)$: algebraic rank = analytic rank.
- A Clay Millennium Prize Problem; proved only for analytic rank ≤ 1 (Gross–Zagier, Kolyvagin), open in general.
- So BSD itself was **found by computer** — the conjecture read off the data, not derived from theory.



$\prod_{p\le P} N_p/p$ vs $\log P$ (computed): rank-1 grows, rank-0 stays bounded.

Birch–Swinnerton-Dyer: the precise statement

For $E/\mathbb{Q}$ with $r = \operatorname{ord}_{s=1} L(E,s)$, the conjecture has two parts:

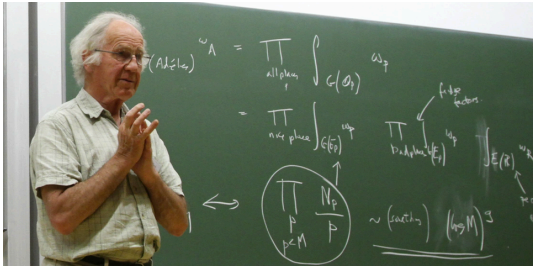
- **Rank:** $r = \operatorname{rank} E(\mathbb{Q})$.
- **Leading coefficient:** $\displaystyle\lim_{s\to 1} \frac{L(E,s)}{(s-1)^r} = \frac{\Omega_E \cdot \operatorname{Reg}(E) \cdot \#Ш(E) \cdot \prod_p c_p}{\left(\#E(\mathbb{Q})_{\mathrm{tors}}\right)^2}.$

Invariants (minimal model):

- Ω_E — real period $\int_{E(\mathbb{R})} |\omega|$ (ω the Néron differential).
- $\operatorname{Reg}(E)$ — regulator: $\det$ of the Néron–Tate height pairing on a basis of $E(\mathbb{Q})/\mathrm{tors}$ ($= 1$ if $r = 0$).
- $\#Ш(E)$ — order of the Tate–Shafarevich group (failure of the local–global principle; conjectured finite).
- c_p — Tamagawa number $[E(\mathbb{Q}_p) : E^0(\mathbb{Q}_p)]$ ($= 1$ at good p); $\prod_p c_p$ finite.
- $\#E(\mathbb{Q})_{\mathrm{tors}}$ — order of the torsion subgroup.

Birch and Swinnerton-Dyer

Working from their EDSAC-2 point counts at Cambridge in the early 1960s, Birch and Swinnerton-Dyer read off the rank–L-function correlation that became the conjecture.



Bryan Birch (b. 1931), British; University of Oxford.
Also known for Birch's theorem.

Sir Peter Swinnerton-Dyer (1927–2018), British; University of Cambridge; 16th Baronet.

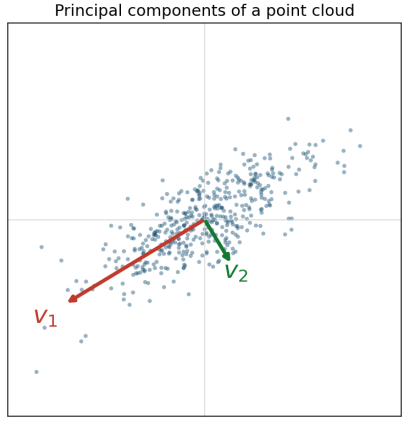
The machine: EDSAC 2

- A vacuum-tube (valve) computer at the Cambridge Mathematical Laboratory, running from **1958** — successor to EDSAC (1949).
- Built under Maurice Wilkes with David Wheeler: the **first computer with a microprogrammed control unit**, and the first bit-slice architecture.
- Birch and Swinnerton-Dyer ran their elliptic-curve point counts on it; the conjecture came out of those runs.
- Its small store and slow speed limited the runs to small-conductor curves.

EDSAC 2 in 1960, Cambridge Mathematical Laboratory. Wikimedia Commons.

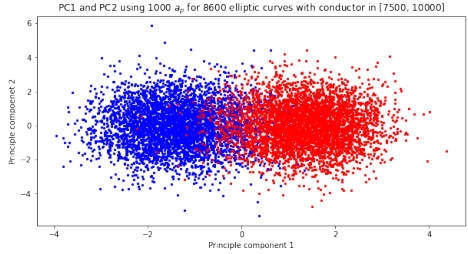
Principal component analysis (PCA)

- An **unsupervised** linear dimensionality-reduction method: orthogonal directions (principal components) ordered by the variance they capture.
- Centre the points; the eigenvectors $v_1, v_2, \dots$ of the covariance $C = \frac{1}{n} X^\top X$ (eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$) are the components — v_1 the most-variance direction, $v_2 \perp v_1$ next, λ_i the variance along v_i (the SVD of X).
- Project onto the top two, $x \mapsto (v_1 \cdot x,\ v_2 \cdot x)$, to plot high-dimensional data in 2-D.
- Unsupervised: uses the data, not the labels, so any clusters are intrinsic.



A machine-learning rank classifier

- The **LMFDB** (L-functions and Modular Forms Database) lists elliptic curves with rank, conductor, and (a_p) — labelled data.
- He, Lee, and Oliver, with Lee's undergraduate Pozdnyakov (2020–22), trained classifiers to predict rank from $(a_{p_1}, \ldots, a_{p_{1000}})$.
- Ranks separate cleanly: logistic regression reaches precision 0.96 on $\{0, 1\}$ and 0.9998 on $\{1, 2\}$; principal component analysis (PCA), a top-variance projection, separates them visually.
- So the a_p carry the rank. *What* in them? Pozdnyakov averaged them and plotted.



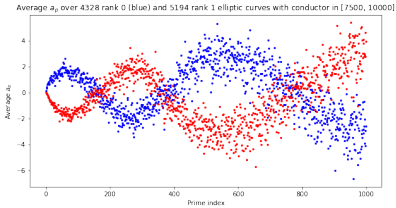
PCA of the a_p vectors, conductor $[7500, 10000]$; ranks separate. He–Lee–Oliver–Pozdnyakov, arXiv:2204.10140.

The murmuration: the averaged statement

- The **conductor** $N(E)$ is built from the bad primes (dividing Δ); it orders curves by size.
- In a band $[N_1, N_2]$, let $\mathcal{E}_r$ be the rank-r curves. Average the n-th Frobenius trace:

$$f_r(n) = \frac{1}{\#\mathcal{E}_r} \sum_{E \in \mathcal{E}_r} a_{p_n}(E).$$

- **Finding (2022):** f_0, f_1 trace two mirrored oscillating curves, amplitude and period growing with n; approximately scale-invariant across bands.
- Named *murmurations* for the look of starling flocks. Empirical: no formula, no proof.



f_0, f_1, conductor $[7500, 10000]$ (arXiv:2204.10140).

A starling murmuration.

The murmuration discoverers

- **Yang-Hui He** (London Institute for Mathematical Sciences) — brought machine learning to these number-theory datasets.
- **Kyu-Hwan Lee** (University of Connecticut) and **Thomas Oliver** (University of Westminster) — set up the rank-classification experiments on L-function data.
- **Alexey Pozdnyakov** (then Lee's undergraduate student) — averaged the a_p by rank and plotted them: the murmuration.

Published in *Experimental Mathematics* 34 (2025) (arXiv:2204.10140).

Yang-Hui He, London Institute for Mathematical Sciences. Wikimedia Commons.

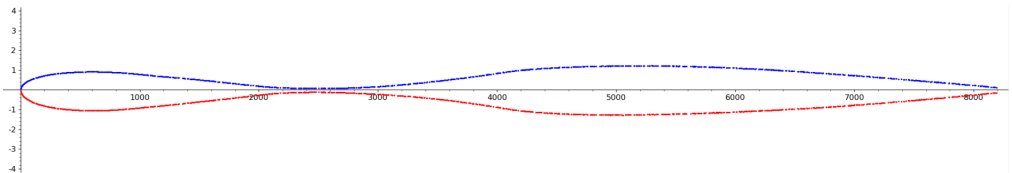
From elliptic curves to modular forms

- Sutherland (MIT) reproduced murmurations on $> 10^9$ curves and in far more general L-functions — the phenomenon is not special to elliptic curves.
- The clean setting is **modular forms**: a holomorphic function on the upper half-plane transforming under $\Gamma_0(N)$ (level N, weight k); a *newform* is a Hecke eigenform genuinely of level N, with Fourier coefficients $a_f(p)$.
- By modularity an elliptic curve *is* a weight-2 newform with the same a_p — so the elliptic-curve murmuration is the weight-2 case.
- Each f has a **root number** $\varepsilon(f) = \pm 1$ (sign of its functional equation), the analogue of rank parity — group forms by $\varepsilon(f)$. Normalized coefficient $\lambda_f(p) = a_f(p)/p^{(k-1)/2}$, $|\lambda_f(p)| \leq 2$ (the analogue of $a_p/\sqrt{p}$).

Sutherland's limit curve

The elliptic-curve murmuration averaged a_p over rank classes and plotted it against the prime index — and was scale-invariant. Sutherland makes that exact:

- Fix a band of levels $N \in [X, cX]$ and one prime P that scales with the level, $P \sim N$.
- **Root-number-weighted average:** multiply each $a_f(P)$ by its sign $\varepsilon(f)$, then average over all weight-k newforms in the band — the signs make the bias add up instead of cancel.
- As $N \to \infty$ this converges to one **continuous curve** depending only on $y = P/N$ (prime size ÷ level size).



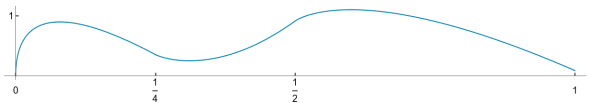
Average of $\sqrt{P}\,\lambda_f(P)\,\varepsilon(f)$ over levels $[2^{13}, 2^{14}]$ vs $y = P/N$. Courtesy of Sutherland, in arXiv:2310.07681.

Zubrilina's theorem (2023): the murmuration density

Average $\sqrt{P}\,\lambda_f(P)\,\varepsilon(f)$ over all weight-k newforms f of square-free level $N \in [X, X+Y]$, with $Y = (1+o(1))X^{1-\delta}$ for a suitable $\delta > 0$; $\lambda_f(P) = a_f(P)/P^{(k-1)/2}$ is the normalized coefficient, $\varepsilon(f) = \pm 1$ the root number, and $y = P/X$. As $X \to \infty$:

$$\frac{\sum_N \sum_f \sqrt{P}\,\lambda_f(P)\,\varepsilon(f)}{\sum_N \sum_f 1} \;=\; \mathcal{M}_k(y) \;+\; (\text{power-saving error}).$$

- $\mathcal{M}_k$ is explicit and continuous: a finite sum of Chebyshev polynomials U_{k-2} with Euler-product coefficients, plus a $\sqrt{y}$ term and (for $k = 2$) a y term.
- Proof: the Eichler–Selberg trace formula turns the average into an average of class numbers. The elliptic-curve murmuration is the case $k = 2$ (arXiv:2310.07681).



The murmuration is now a family of theorems

Zubrilina's weight-k density was the first proof. The phenomenon has since been established for further families:

- **Dirichlet characters** — Lee, Oliver, Pozdnyakov (arXiv:2307.00256).
- **Weight aspect** — Bober, Booker, Lee, Lowry-Duda; level fixed, weight $\to \infty$ (arXiv:2310.07746).
- **Maass forms** — the non-holomorphic case (arXiv:2409.00765).
- **Ordered by height** — the same curves ordered by naive height (arXiv:2504.12295).
- **Function fields** — the function-field analogue (arXiv:2603.13802).

Computing the rank vs predicting it

- **Certified** rank: descent (e.g. `mwrank`, Sage) — correct when it finishes, but it can stall when the Tate–Shafarevich group is large; no algorithm is proven to terminate for every curve.
- **Predicting** the rank from the a_p is easier — and the a_p determine it, through the explicit formula

$$-\frac{L_E'(s)}{L_E(s)} = \sum_{n\geq 1} \frac{c_n \Lambda(n)}{n^s} = -\frac{r}{s-1} + \cdots,$$

  where Λ is the von Mangoldt function, $c_p = a_p$, and $r = \operatorname{ord}_{s=1} L(E,s)$ is the analytic rank; by BSD it equals the algebraic rank. The minus sign is the bias: higher rank pushes the a_p sums more negative.
- So a weighted prime sum of the a_p estimates r — prediction, not proof.

Mestre–Nagao sums

Explicit prime sums that estimate the analytic rank (Mestre 1982, Nagao 1992; the list S_0–S_6 collected in arXiv:2207.06699). The basic one:

$$S_0(B) = \frac{1}{\log B} \sum_{\substack{p<B \\ \text{good}}} \frac{a_p(E)\,\log p}{p}.$$

- **Kim–Murty:** under the Riemann Hypothesis for L_E, if the limit exists, $\lim_{B\to\infty} S_0(B) = -r + \frac{1}{2}$.
- Variants: Nagao's S_3 (logarithmic derivative of the partial Euler product at $s = 1$, large for large rank); Bober's S_6 (from the explicit formula, $\to r$ under RH, but infeasible beyond a small range).
- A sum returns a *number*; reading a rank off it needs a decision rule and a conductor-dependent cutoff.

Learning the rank: trees and a CNN

- **Gradient-boosted trees** (Alessandretti–Baronchelli–He, 2019, arXiv:1911.02008): 2.5M Cremona curves; an ensemble of decision trees finds correlations among rank, Weierstrass coefficients, conductor, regulator, Tamagawa number, and the order of the Tate–Shafarevich group.
- **A 1-D CNN** (Kazalicki–Vlah, 2022, arXiv:2207.06699) learns the rank from a curve's a_p sequence.
- On the LMFDB it beats every Mestre–Nagao sum: Matthews correlation coefficient (a balanced score, max 1) 0.9958 — only 0.25% misclassified — vs 0.87 for the best single sum. A net fed all sums at once beats any one sum.
- On a harder set (conductor up to 10^{30}, rank to 10) accuracy drops; very large conductors need more a_p.

The CNN rank classifier: input, output, product

- **Input.** One curve becomes a $3 \times \pi(N)$ matrix; each column is a prime $p < N$ ($N = 10^3, 10^4, 10^5$):
  - row 1: $a_p/\sqrt{p} \in [-2, 2]$ (Hasse) — this row carries the rank;
  - row 2: the normalized conductor $\log N_E / \log N_{\max}$;
  - row 3: the prime's position in the sequence.
- **Output.** A probability for each rank class $0, 1, 2, \ldots$; the prediction is the largest. A classification, not a single number.
- **Training.** Supervised on LMFDB curves whose rank is already known (certified by descent); minimise the cross-entropy loss by back-propagation. The 1-D convolution slides one learned filter along the prime axis.
- **Product.** A fixed-weight function f_θ: (a_p matrix) $\mapsto$ rank — a predictor with no certificate. It estimates the analytic rank (= algebraic under BSD); certifying the rank still needs descent.

The 1-D convolution: a sliding weighted sum

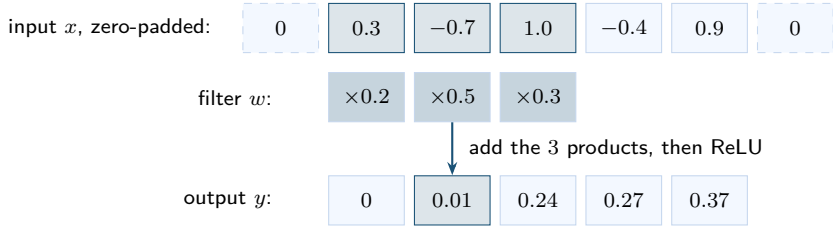
First layer, kernel size 17. Output channel $j \in \{1, \ldots, 64\}$ at prime position i:

$$y_j[i] \;=\; \mathrm{ReLU}\Big(b_j + \sum_{c=1}^{3} \sum_{k=-8}^{8} W_j[c,k]\; x_c[i+k] \Big), \qquad \mathrm{ReLU}(t) = \max(t, 0).$$

- **Window:** the 17 columns centered at position i, all 3 channels — a 3×17 block of the input.
- **Filter:** W_j, a 3×17 array of weights, plus a bias b_j — 52 learned numbers.
- Multiply window and filter entry by entry, add the 51 products, add b_j, clip negatives to 0: one number — entry i of output row j.
- The same (W_j, b_j) is used at every $i = 1, \ldots, 1229$ — the filter slides. 64 filters give the 64 output rows.
- At the two ends the window sticks out; missing entries are set to 0 (zero padding), so the length is kept.

One convolution by hand: kernel size 3, one channel

Filter $w = (0.2,\ 0.5,\ 0.3)$, bias $b = 0$: $\quad y[i] = \mathrm{ReLU}\big(0.2\,x[i{-}1] + 0.5\,x[i] + 0.3\,x[i{+}1]\big)$.



- $y[2] = \mathrm{ReLU}\big(0.2(0.3) + 0.5(-0.7) + 0.3(1.0)\big) = \mathrm{ReLU}(0.01) = 0.01.$
- Slide one step and repeat: $y[3] = 0.24$, $y[4] = 0.27$, $y[5] = 0.37$.
- Left edge: $y[1] = \mathrm{ReLU}\big(0.2(0) + 0.5(0.3) + 0.3(-0.7)\big) = \mathrm{ReLU}(-0.06) = 0.$
- The same three weights at every position; a fully-connected layer would have separate weights for each position.

Stride 2, the later layers, and the parameter count

- **Stride 2** (the reducing layers): the same formula, computed only at every other position $i = 1, 3, 5, \ldots$ — the length halves.
- **From the second layer on:** the input has 64 channels, so each filter is a 64×17 block — one output entry is a sum of $64 \cdot 17 = 1088$ products plus a bias. Still 64 filters per layer. (Once the sequence is shorter than 17, the kernel is capped at the sequence length.)
- **Parameter count:** first layer $64 \cdot 3 \cdot 17 + 64 = 3{,}328$; a full-length later layer $64 \cdot 64 \cdot 17 + 64 = 69{,}696$; with the shrinking kernels near the end, the whole model holds $596{,}293 \approx 6 \times 10^5$ learned numbers, all set by gradient descent on the cross-entropy loss.
- Without its ReLU, a convolution layer is a linear map that is translation-equivariant: shift the input along the prime axis and the output shifts the same way.
- After the 11 halvings, each of the final 64 numbers depends on all 1229 primes — 64 learned whole-sequence statistics; the dense layer is a linear classifier on them. A Mestre–Nagao sum is one hand-designed statistic of the same kind.

The CNN design (Kazalicki–Vlah): four layer types

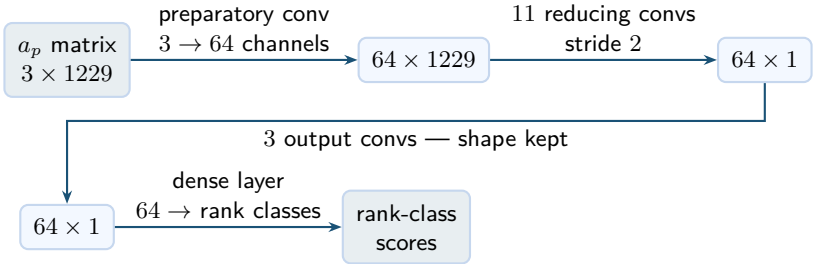
Design rationale: a Mestre–Nagao sum applies the same rule to every a_p — a shared weighting — and a 1-D convolution is exactly such a shared filter slid along the primes. Write $C \times L$ for C channels of length L. The network stacks 1-D convolutions in four groups:

- **Preparatory** (1 layer): $3 \times L \to 64 \times L$ — widens the 3 input rows to 64 channels; the length is unchanged.
- **Input** (L_1 layers): $64 \times L \to 64 \times L$ — same shape; extra processing steps before any shortening.
- **Reducing** (L_2 layers): stride 2 — each halves the length; L_2 is set so the length reaches exactly 1.
- **Output** (L_3 layers): $64 \times 1 \to 64 \times 1$ — same shape; extra processing steps after the shortening.
- **Final dense layer:** the last 64 numbers $\to$ one score per rank class; the largest score is the predicted rank.

Every convolution is followed by ReLU and batch normalization; one kernel size KS throughout, capped at the sequence length once the sequence is shorter than the kernel. The whole net is fixed by (L_1, L_2, L_3, KS), chosen by Bayesian optimization over 500 trained variants.

The LMFDB $p < 10^4$ model $(0, 11, 3, 17)$, layer by layer

$\pi(10^4) = 1229$ primes below 10^4, so one curve enters as a 3×1229 matrix; $L_1 = 0$, so this model has no input layers.



- $2^{10} < 1229 \le 2^{11}$: eleven halvings take the length $1229 \to \cdots \to 1$ (the code sets $L_2 = \lceil \log_2 \pi(N) \rceil$).
- Each convolution: `Conv1d` with zero padding $\lfloor \text{kernel}/2 \rfloor$, then ReLU, then batch norm.
- Training: class-weighted cross-entropy, Adam, one-cycle schedule, 40 epochs, batch size 2048.

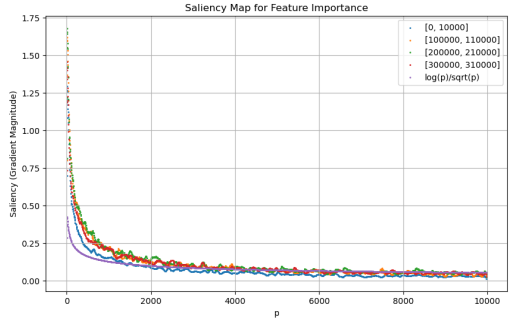
The $p < 10^4$ network, every layer (from the published code)

layer	operation	kernel	stride	output shape
input	$a_p/\sqrt{p}$, conductor, position	—	—	3×1229
preparatory	conv $3 \to 64$	17	1	64×1229
R_1	conv $64 \to 64$	17	2	64×615
R_2	conv $64 \to 64$	17	2	64×308
R_3	conv $64 \to 64$	17	2	64×154
R_4	conv $64 \to 64$	17	2	64×77
R_5	conv $64 \to 64$	17	2	64×39
R_6	conv $64 \to 64$	17	2	64×20
R_7	conv $64 \to 64$	17	2	64×10
R_8	conv $64 \to 64$	11	2	64×5
R_9	conv $64 \to 64$	5	2	64×3
R_{10}	conv $64 \to 64$	3	2	64×2
R_{11}	conv $64 \to 64$	3	2	64×1
O_1–O_3	conv $64 \to 64$ ($\times 3$)	1	1	64×1
head	flatten + linear $64 \to 5$	—	—	5 scores (ranks 0–4)

Every conv row = `Conv1d` $\to$ ReLU $\to$ batch norm. The kernel is capped at the sequence length, so it shrinks near the end; the O layers are 64×64 channel-mixing maps. Parameters: $594{,}048 + 1{,}920 + 325 = 596{,}293$.

The network's saliency: murmuration early, Mestre–Nagao late

- Saliency w_p — the gradient of the output with respect to each a_p — shows what the CNN relies on.
- **Early** in training the rank-0-vs-rank-1 saliency shows the *murmuration* oscillation.
- **Later** the rank-0 saliency flattens and weight moves to small primes, matching the Mestre–Nagao weighting $\log p/\sqrt{p}$.
- Murmurations matter more for small conductors. Test accuracy 99.85% on $[0, 10^4]$, 98.57% on $[3\times10^5, +10^4]$.



Trained CNN saliency across four conductor ranges vs the Mestre–Nagao weighting $\log p/\sqrt{p}$ (purple).
arXiv:2603.17681.